Q1: k-Means Clustering in High Dimensions

COL761 Assignment 2, 2026 BITBLASTING

(a) Selection of k

For each $k \in \{1, \ldots, 15\}$ we minimise the k-means objective (WCSS) via Lloyd's algorithm with 10 random restarts and record the inertia $I(k) = \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$. Features are z-normalised before fitting to prevent scale dominance.

Method. The inertia curve is monotonically decreasing; the *elbow* marks where marginal gain drops sharply. We formalise this via the *normalised acceleration ratio*:

$$r(k) = \frac{\Delta^2 \tilde{I}(k)}{|\tilde{I}(k)| + \varepsilon}, \qquad \tilde{I}(k) = \frac{I(k) - I_{\min}}{I_{\max} - I_{\min}},$$

where Δ^2 denotes the second finite difference. Dividing by $|\tilde{I}(k)|$ normalises the curvature relative to the current inertia magnitude, avoiding false positives in the steep, low-k region. The selected k is $\operatorname{argmax}_k r(k)$.

The resulting elbow and the supplementary silhouette scores are shown in Figure 1. Both criteria agree on the selected value.

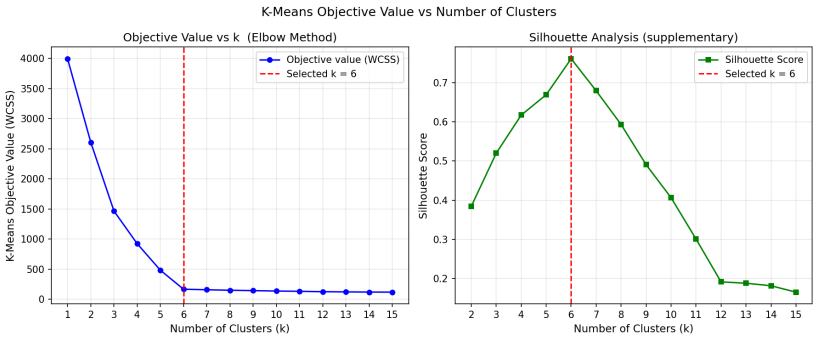

Figure 1: Left: WCSS vs. k with the selected k (red dashed line). Right: Silhouette score vs. k (supplementary).

(b) Structural and Statistical Assumptions

1. **Spherical, isotropic clusters.** k-means minimises squared Euclidean distances, implicitly assuming clusters are convex and roughly equal-radius hyperballs.
2. **Equal cluster sizes.** The Voronoi partition favours balanced clusters; small, dense clusters are absorbed by large ones.
3. **Gaussian generative model.** WCSS is the MLE objective for a mixture of isotropic Gaussians $\mathcal{N}(\mu_j, \sigma^2 I)$ with equal mixing weights and shared variance.
4. **Euclidean metric appropriateness.** In high dimensions, $\|x - \mu\|$ concentrates sharply around $\sqrt{d}\,\sigma$ for any μ (Lévy's lemma / measure concentration [1]), so inter- vs. intra-cluster distance ratios shrink, degrading separation.
5. k **known a priori.** The model has no mechanism to infer k from data; it must be supplied externally.

(c) Limitations of the Model Selection Procedure

- **Curse of dimensionality.** In high d, the volume of a ball of radius r satisfies $V_d(r) \propto r^d/\Gamma(d/2+1) \to 0$ for fixed r as $d \to \infty$ [1]. Distances concentrate, making the elbow extremely flat and hard to detect reliably.
- **Sensitivity to normalisation.** The elbow location can shift with feature scaling or normalisation choices.
- **Local optima.** Lloyd's algorithm is not guaranteed to find the global WCSS minimum; different restarts may yield different inertias, adding noise to the elbow curve.
- **No probabilistic model.** WCSS provides no likelihood, so formal model comparison (AIC/BIC) is unavailable without additional assumptions.

(d) Methodological Improvements and Alternatives

- **BIC/AIC on GMM.** Replace k-means with a full Gaussian Mixture Model and select k by BIC, which penalises model complexity explicitly [3].

- **Gap statistic.** Compare observed $\log I(k)$ against its expectation under a uniform null reference distribution; the optimal k is the smallest k where the gap $G(k) = E[\log I_{\text{ref}}(k)] - \log I(k)$ exceeds $G(k+1) - s_{k+1}$ [2].
- **Stability-based selection.** Cluster on random subsamples and measure clustering stability (e.g. adjusted Rand index); the most stable k is preferred [4].
- **Dimensionality reduction pre-processing.** Applying PCA or UMAP before clustering mitigates distance concentration and often sharpens the elbow.

(e) Failure Modes of k-Means

k-Means is inappropriate in the following well-known settings.

Non-convex / manifold-shaped clusters (e.g. two interlocking rings): the Voronoi partition cannot recover non-convex cluster boundaries. *Detection:* fit a k-nearest-neighbour graph and check whether the resulting connectivity components match the k-means labels; persistent disagreement signals non-convexity. *Alternative:* spectral clustering or DBSCAN.

Clusters of very different sizes or densities: large clusters "steal" points from small dense ones to lower WCSS. *Detection:* compare cluster cardinalities and intra-cluster variances; high imbalance suggests misfit. *Alternative:* GMM with full covariance or DBSCAN.

High-dimensional data with distance concentration: as d grows, the ratio $\max_i \|x - x_i\| / \min_i \|x - x_i\| \to 1$ [5], so nearest-centroid assignment becomes nearly random. *Detection:* compute the coefficient of variation of pairwise distances; values near 0 signal concentration. *Mitigation:* use projected features ($d' \ll d$) or subspace clustering.

Categorical or mixed-type features: squared Euclidean distance is semantically meaningless for discrete or ordinal variables. *Alternative:* k-modes or k-prototypes.

References

[1] A. Blum, J. Hopcroft, R. Kannan. *Foundations of Data Science*. Cambridge University Press, 2020.

[2] R. Tibshirani, G. Walther, T. Hastie. Estimating the number of clusters via the gap statistic. *JRSS-B*, 63(2):411–423, 2001.

[3] G. McLachlan, D. Peel. *Finite Mixture Models*. Wiley, 2000.

[4] S. Ben-David, U. von Luxburg, D. Pál. A sober look at clustering stability. *COLT*, 2006.

[5] K. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft. When is "nearest neighbor" meaningful? *ICDT*, 1999.